

Dental AI

An Intelligent System for Automated Landmark Detection, Forensic Age Estimation, and AI-Assisted Interpretation of Panoramic Dental Radiographs

Abstract

Dental radiographs, particularly panoramic radiographs (Orthopantomograms, OPGs), are indispensable tools in diagnosis, treatment planning, orthodontics, and forensic investigations. However, manual identification of anatomical landmarks and estimation of dental age remain expertise-intensive, time-consuming, and prone to inter-observer variability. This project presents Dental AI, an intelligent dental analysis system that integrates Artificial Intelligence, Deep Learning, and Large Language Models (LLMs) to assist dental professionals, students, and forensic experts in interpreting panoramic dental radiographs.

The primary objective of the proposed system is the automated detection of dental anatomical landmarks from panoramic X-ray images. Using advanced deep learning-based landmark detection techniques, the system accurately identifies clinically significant structures, including the mandibular canal, mental foramen, maxillary sinus, condyle, coronoid process, cervical vertebrae, and other anatomical features. These detected landmarks provide valuable support for orthodontic treatment planning, implant placement, surgical procedures, and comprehensive anatomical assessment.

A key innovation of the project is its AI-driven forensic age estimation module, which analyzes the developmental stages and morphological characteristics of dental structures to estimate an individual's biological age from panoramic radiographs. This capability has significant applications in forensic odontology, disaster victim identification, legal investigations, and age verification.

To enhance usability and accessibility, the system incorporates an LLM-powered conversational assistant that enables users to interact with the platform using natural language. This chatbot explains detected landmarks, interprets AI-generated results, answers questions on dental anatomy and radiology, and offers educational guidance, making the system valuable for both clinical practice and dental education.

By unifying computer vision, forensic dentistry, and generative AI within a single platform, Dental AI demonstrates how intelligent technologies can improve diagnostic accuracy, streamline clinical workflows, support forensic investigations, and create an interactive learning environment. The proposed system represents a significant step towards the adoption of AI-assisted decision support in modern dentistry.

Keywords: Panoramic Radiography (OPG), Anatomical Landmark Detection, Forensic Age Estimation, Deep Learning, Computer Vision, Large Language Models, AI-Assisted Dentistry, Dental Radiology, Forensic Odontology.